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Complexes between hypohalous acids and phosphine derivatives. Pnicogen bond versus halogen bond versus hydrogen bond

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Abstract

The complexes of HOBr:PH_2Y (Y = H, F, Cl, Br, CH_3, NH_2, OH, and NO_2), HOCl:PH_2F, and HOI:PH_2F have been investigated with ab initio calculations at the MP2/aug-cc-pVTZ level. Four types of structures (**1**, **2**, **3a**, and **3b**) were observed for these complexes. **1** is stabilized by an O···P pnicogen bond, **2** by a P···X halogen bond, **3a** by a H···P hydrogen bond and a P···X pnicogen bond, and **3b** by H···P and H···Br hydrogen bonds. Their relative stability is related to the halogen X of HOX and the substituent Y of PH_2Y. These structures can compete with interaction energy of -10.22 ~ -29.40 kJ/mol. The H-O stretch vibration shows a small red shift in **1**, a small irregular shift in **2**, but a prominent red shift in **3a** and **3b**. The X-O stretch vibration exhibits a smaller red shift in **1**, a larger red shift in **2**, but an insignificant blue shift in **3a** and **3b**. The P-Y stretch vibration displays a red shift in **1** but a blue shift in **2**, **3a**, and **3b**. The formation mechanism, stability, and properties of these structures have been analyzed with molecular electrostatic potentials, orbital interactions, and non-covalent interaction index.

Keywords: Hydrogen bonds; Halogen bonds; Pnicogen bonds; Hypohalous acids; Phosphine derivatives

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1. Introduction

Noncovalent interactions play an extensive and important role in fields of physics, chemistry, and biology [1,2]. With progress of the study on noncovalent interactions, different types of interactions such as hydrogen bond, halogen bond, and pnicogen bond have been proposed [3-5]. Undoubtedly, hydrogen bond is still one of the most important noncovalent interactions, although lots of study has been performed on it. In 2011, a modern evidence-based definition of hydrogen bond was recommended [6]: *The hydrogen bond is an attractive interaction between a hydrogen atom from a molecule or a molecular fragment X–H in which X is more electronegative than H, and an atom or a group of atoms in the same or a different molecule, in which there is evidence of bond formation*. Halogen bond shares some similarities in structures, properties, strengths, and applications with hydrogen bond, thus in recent years much attention has been paid to it [7-12], due to its potential applications in molecular recognition [13], crystal materials [14], and biological systems [15]. IUPAC recommendation in 2013 for halogen bond is: *A halogen bond occurs when there is evidence of a net attractive interaction between an electrophilic region associated with a halogen atom in a molecular entity and a nucleophilic region in another, or the same, molecular entity* [16].

Pnicogen bond was recognized as a new and important noncovalent interaction [17-33]. This interaction is analogous to halogen bond since the formation of both interactions is related to the σ-hole on the outer surface of bonded pnicogen and halogen atoms [5]. NBO analyses showed that the lone pair electrons in Lewis bases transfer into the unoccupied orbital of P-X bond directed away from the P atom in pnicogen-bonded complexes involved with PH_2X (X = H, F, Cl, CH_3, NH_2, CF_3, HO, and NO_2) [19]. The electron-withdrawing

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substituent exerts a strengthening influence on pnicogen bonds [19]; however, the electron-donating substituent NH_2 also brings out a similar enhancing effect, although its influence is smaller. Pnicogen bond was thought to be a new molecular linker in supermolecular chemisty [34] and to be relevant in biological systems [35].

Hypohalous acids as oxidants are involved in the seasonal depletion of the ozone layer in the stratosphere [36-38] and pathophysiological processes [39]. They are a polyfunctional molecule with an acidic proton, an alkaline oxygen, and a halogen with anisotropic electrostatic potentials on its surface, thus hypohalous acids form different types of complexes with themselves [40-42] or other molecules [43-55]. These clusters are helpful for the understanding the role of hypohalous acids in atmospheric chemistry and biological systems.

In the present study, we analyze the complexes of hypobromous acid (HOBr) and phosphine derivatives (PH_2Y, Y = H, F, Cl, Br, CH_3, NH_2, OH, and NO_2) using theoretical calculations. For comparison, the complexes of HOCl:PH_2F and HOI:PH_2F are also studied. Theoretical methods are feasible and necessary for studying these complexes because of chemical instability of hypohalous acids, which makes a difficulty in studying them with experimental methods. Both components have a dual character as a Lewis acid and base, thus it is interesting to know what types of interactions are formed between them, what are the structures of the equilibrium complexes found at these minima, and how do they compare energetically. To characterize these complexes, we analyze the nature of their intermolecular bonds and charge densities.

2. Computational details

The structures of complexes and their isolated monomers were first optimized at

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second-order Møller–Plesset perturbation theory (MP2) with the basis set aug-cc-pVDZ for the atoms except iodine, for which the basis set aug-cc-pVDZ-PP was used to take into account for relativistic effects. All structures were then optimized at the MP2/aug-cc-pVTZ(PP) level to obtain more reliable results. Finally harmonic vibrational frequency calculations were carried out at the MP2/aug-cc-pVTZ(PP) level to confirm that the structures obtained correspond to energetic minima. All calculations were performed using the Gaussian09 program [56]. The interaction energy, a difference between the total energy of the complex and the energy sum of the respective monomers, was corrected for the basis set superposition error (BSSE) using the counterpoise method of Boys and Bernardi [57].

Molecular electrostatic potentials (MEPs) on the 0.001 electrons/bohr^3 contour of electronic density were calculated with the Wave Function Analysis-Surface Analysis Suite (WFA-SAS) program [58]. Natural bond orbital (NBO) analyses were obtained with NBO 3.1 version [59] implemented in Gaussian 09 to provide an insight into the bonding nature of these complexes. The atoms in molecules (AIM) analyses were carried out with AIM2000 package [60] to obtain bond topological properties including electron density and Laplacian. The non-covalent interaction (NCI) index was analyzed with the Multiwfn program [61] and its plot was graphed with VMD program [62].

3. Results and discussion

3.1. MEPs of isolated monomers

MEP is a useful tool for identifying basic and acidic centers for intermolecular interactions [63], thus the MEP maps of PH_2F and HOCl as the respective representatives of phosphine derivatives and hypohalous acids are plotted in Fig. 1, and the MEP maps of all

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monomers are given in Fig. S1. Here we are interested in the MEPs on the P atom of PH_2F. There is a negative MEP region associated with the lone pair on P and a positive MEP region (σ-hole) centered on the extension of the P-Y bond, which is consistent with the previous results [19,28,29]. One can see from Table 1 that when Y of PH_2Y varies from F through Cl to Br, the negative MEP value on the P atom becomes less negative and the positive one at the σ-hole decreases. The negative MEP value on the P atom becomes more negative in the order OH < H < NH_2 < CH_3, while the positive MEP at the σ-hole increases in the order CH_3 < H < NH_2 < OH. This order is the same as that in Ref. 19. The MK electrostatic potential charge on the CH_3 group of PH_2CH_3 is 0.059e, indicating the methyl group is electron-donating, which is responsible for the increase of negative MEP value on the P atom and the decrease of positive MEP at the σ-hole in PH_2CH_3 relative to PH_3. This charge is -0.062e and -0.061e for the NH_2 and OH groups, respectively, indicating both groups are electron-withdrawing in PH_2NH_2 and PH_2OH, which is also consistent with the fact that the N and O atoms have greater electronegativity than the P atom. The electron-withdrawing ability of OH group is greater than that of NH_2, which is responsible for the bigger positive σ-hole on the P atom of PH_2OH. The strong electron-withdrawing nitryl group results in a great increase of the positive MEP at the σ-hole and even makes the MEP of the lone pair on P become positive.

For the MEP of HOCl, it is positive on the H atom, negative on the O atom, and anisotropic on the Cl atomic surface, in which a σ-hole is found on the outer surface of bonded Cl atom. This feature was also observed for other hypohalous acids [49-55]. With the increase of X atomic number in HOX, the positive MEP on the H atom decreases, the positive MEP of the σ-hole on the X atom increases, the negative MEP on the X atom decreases, and

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the negative MEP on the O atom becomes more negative. More negative MEPS are associated with stronger electrophilic sites, while more positive MEPs are indicative of strong nucleophilic sites.

3.2. Structures and energies

According to the MEPs on the P atom of PH_2Y, it is inferred that P can act as a dual role of Lewis acid and base to interact with the O and H or X of HOX, respectively. Fig. 2 is a schematic drawing that illustrates the representative complexes (**1**, **2**, **3a**, and **3b**) formed between hypohalous acids and phosphine derivatives. Here we are interested in the interaction between the P of PH_2Y and the H, X, and O atoms of HOX, although there are also other interactions between the Y of PH_2Y and HOX. The optimized structures of all complexes are given in Fig. S2. In **1**, a pnicogen bond occurs between the σ-hole on the P atom of PH_2Y and the lone pair on the O atom of HOX, which is characterized by the presence of a bond critical point connecting the P atom of PH_2Y with the O atom of HOX (Fig. S3). The O-X bond of HOX is in a *cis* position with one P-H bond of PH_2Y, while the O-H bond of HOX is in a *trans* position with the other P-H bond of PH_2Y. The negative electrostatic potential on the halogen surface approaches the positive electrostatic potential on the hydrogen surface in the former. The latter is attributed to the repulsion between the two positively charge hydrogen atoms. In **2**, a halogen bond is formed between the σ-hole on the X atom of HOX and the lone pair on the P atom of PH_2Y, which is characterized by the presence of a bond critical point connecting the P atom of PH_2Y with the X atom of HOX (Fig. S3). The distant O-H bond of HOBr can display three different orientations with the P-Y bond of PH_2Y in **2** with a cis-configuration for Y = NO_2, trans-configuration for Y = CH_3, and cross configuration for Y

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= H, F, Cl, Br, NH_2, and OH. There is an imaginary frequency in **2** of HOBr:PH_2Br if the O-H bond of HOBr has a cis-configuration with the P-Br bond of PH_2Br. For **2** of HOBr:PH_2Cl, the cross configuration is more stable than the cis-configuration because the binding distance is 2.839 Å and 2.848 Å, respectively. However, only the most stable structure of **2** is studied here. In **3a** and **3b**, the positively charge H atom of HOX is introduced into the lone pair on the P atom of PH_2Y. Simultaneously, the X atom of HOX attacks the σ-hole on the P atom of PH_2Y in **3a** and the acidic site of Y group in **3b**. Both types of interactions in **3** are characterized by the presence of the respective bond critical points in Fig. S3 and their cyclic structures are further characterized by the presence of a ring critical point (Fig. S3). **3a** is obtained for all complexes, while **3b** is found only for Y = CH_3, NH_2, and OH. **1** was not obtained for the complexes of HOBr:PH_2Y (Y = H, CH_3, NH_2, and OH) since it was changed to be **3a** for Y = H and **3b** for Y = CH_3, NH_2, and OH in the optimization.

Table 2 presents the interaction energies corrected for BSSE in these complexes. For HOX:PH_2F (X = Cl, Br, and I), with the increase of X atomic number, the interaction energy becomes more negative in all structures, and its dependence on the nature of X is larger in **2** but small in **1** and **3a**. The interaction energies in **1** and **2** respectively show a consistent change with the negative electrostatic potential on the O atom and positive electrostatic potential on the X atom of HOX, while that in 3a is inconsistent with the positive electrostatic potential on the H atom and negative electrostatic potential on the X atom of HOX. The HOCl complexes become more stable in order of **2** < **1** < **3a**, the stabling order is **1** < **2** < **3a** for the HOBr complexes, and the HOI complexes is more stable in order of **1** < **3a** < **2**. For HOBr:PH_2Y (Y =F, Cl, and Br), with the increase of Y atomic number, the interactions in **1**

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and **3a** become weaker, whereas the interaction strength in **2** has a small change. The interaction energy in **1** has a consistent change with the positive electrostatic potential on the P atom of PH_2Y. Such results are like those in the pnicogen-bonded complexes of PH_2Y-NH_3 [19]. The complexes of HOBr:PH_2Y (Y =F, Cl, and Br) become more stable in order of **1** < **2** < **3a**. For HOBr:PH_2Y (Y = H, CH_3, NH_2, OH, and NO_2), the substituents of H, CH_3, NH_2, and OH make **1** absent, while NO_2 make it stronger. The substituents of CH_3, NH_2, and OH bring out a stabilizing effect on **2** and **3a** relative to the F group, while they become unstable due to the NO_2 group. **2** becomes more stable in order of H < OH < CH_3 ≈ NH_2, which shows some inconsistence with the negative electrostatic potential on the P atom in PH_2Y. Interestingly, the enhancing effect of CH_3 on the halogen bond is similar to that of NH_2. The interaction energy of halogen bond in HOBr:PH_2CH_3 increases by -10.17 kJ/mol due to the methyl substitution relative to HOBr:PH_2H. This enhancing effect is larger than that in HCCBr-$NCCH_3$ complex [63]. **3b** is more stable than **3a** for the complexes of HOBr:PH_2CH_3, HOBr:PH_2NH_2, and HOBr:PH_2OH.

The binding distance is a good makeup for the interaction energy to estimate the interaction strength. Table 3 presents the binding distances in all complexes. The P···O distance (R_1) in **1** shows a consistent change with the interaction energy in the complexes except HOBr:PH_2NO_2, which has a larger binding distance but more negative interaction energy than HOBr:PH_2F. The van der Waals attractive interaction between the H atom of PH_2Y and the Br atom of HOBr may be responsible for this difference in the interaction energy because the H···Br distance is shorter in HOBr:PH_2NO_2 than that in HOBr:PH_2F. A consistent change is not found between the P···X distance (R_2) and the interaction energy in **2**

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because of the larger difference in configuration. The strength of hydrogen bond in **3a** can be better estimated with the P···H distance (R_3). With the increase of X atomic number in HOX:PH_2F (X = Cl, Br, I), R_3 becomes longer, corresponding to a weaker hydrogen bond, which is consistent with the positive electrostatic potential on the H atom of HOX. Similar results were reported in other hydrogen-bonded complexes of HOX [43-55]. For HOBr:PH_2Y, CH_3 and NH_2 groups make R_3 shorter, while other substituents make it longer. Interestingly, CH_3 results in a shorter R_3 than NH_2. This can be explained with the negative electrostatic potential on the P atom of PH_2Y. The P···H distance is equal in **3a** and **3b** for HOBr:PH_2CH_3, but it is larger in **3b** than that in **3a** for HOBr:PH_2NH_2 and HOBr:PH_2OH, indicative of a stronger hydrogen bond in **3b**. R_6 in **3b** is the distance between the H atom of Y group in PH_2Y and the Br atom of HOBr, and it becomes shorter in order of Y = CH_3 < NH_2 < OH, which is consistent with the acidity of the proton in these groups.

Changes of H-O, X-O, and P-Y bond lengths and bond angles in the complexes are collected in Table S1. In **1**, the H-O, X-O, and P-Y bonds are elongated, and the P-F bond elongation in HOX:PH_2F (X = Cl, Br, I) is related to the interaction strength. In **2**, the associated X-O bond is greatly lengthened, the distant O-H bond has a negligible change, and the P-Y bond has a great contraction. In **3a** and **3b**, the H-O bond is elongated, while the X-O and P-Y bonds are shortened. Furthermore, **3b** shows a bigger change of bond length than **3a**. The H-O bond elongation in **3** is smaller than that of X-O bond in **2**, while the X-O bond contraction in **3** is greater than that of H-O bond in **2**.

The angle (θ_1) in **1** is in a range of 162-168°, which is consistent with the previous conclusion that linear alignment is favorable for the stability of pnicogen-bonded complexes

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[29]. The O-X···P angle (θ_2) in **2** approaches linearity with a value of 165-178°. This orientation is mainly determined by the σ-hole on the halogen atom surface although the repulsion force between the lone pair electrons on the P and X atoms is also important [64]. In contrast, the O-H···P angle (θ_3) in **3a** and **3b** is usually much smaller, due to the cyclic structure.

3.3. Frequency shifts

Table 4 presents the frequency shifts of H-O, X-O, and P-Y stretch vibrations. They all show a red shift in **1** and the red shift is greater for the H-O stretch vibration than for the X-O one due to the lighter mass of hydrogen atom, although the X-O bond has a larger elongation. For the halogen-bonded complexes, the associated X-O stretch vibration displays an observed red shift, while the distant H-O one has a small irregular shift. Such shifts were reported in the previous study for the halogen bonded complexes of hypohalous acids [43-55]. Additionally, the P-Y stretch vibration also shows a blue shift in **2**. The H-O stretch vibration in **3a** and **3b** show a big red shift, while the X-O and P-Y ones show a blue shift.

3.4. NBO analyses

The interactions in these complexes were analyzed with orbital interactions using NBO theory. The orbital interactions (Fig. 3) are respectively $LP_O \rightarrow BD^*_{P-Y}$ and $LP_P \rightarrow BD^*_{X-O}$ in **1**, $LP_P \rightarrow BD^*_{X-O}$ and $LP_X \rightarrow BD^*_{P-Y}$ in **2**, $LP_P \rightarrow BD^*_{H-O}$ and $LP_X \rightarrow BD^*_{P-Y}$ in **3a** for $HOBr:PH_2Br$. They are estimated with second-order perturbation energy (*E*) as given in Table 5. For **1**, the *E* value due to the $LP_O \rightarrow BD^*_{P-Y}$ orbital interaction is much larger than that of $LP_P \rightarrow BD^*_{X-O}$, indicating the former orbital interaction is important for the stability of pnicogen-bonded complexes [19]. This orbital interaction is also responsible for the P-Y bond

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elongation in **1**. For **2**, the *E* value due to the $LP_X \rightarrow BD^*_{P-Y}$ orbital interaction is far smaller than that of $LP_P \rightarrow BD^*_{X-O}$, thus its contribution can be negligible. The interaction energy in **2** shows a linear relationship with the *E* value due to the $LP_P \rightarrow BD^*_{X-O}$ orbital interaction. The $LP_P \rightarrow BD^*_{X-O}$ orbital interaction in **2** is much stronger than the $LP_O \rightarrow BD^*_{P-Y}$ orbital interaction in **1** for all complexes. This orbital interaction has contribution to the X-O bond elongation in **2**. For **3a**, the $LP_P \rightarrow BD^*_{H-O}$ orbital interaction is stronger than the $LP_X \rightarrow BD^*_{P-Y}$ one in the complexes of HOBr:PH_2Y (Y = H, CH_3, NH_2, and OH) and HOCl:PH_2F, while the former is weaker than the latter in other complexes. The complexes of HOBr:PH_2H, HOBr:PH_2CH_3, and HOBr:PH_2NH_2 in **3a** are mainly stabilized by the hydrogen bond, while both hydrogen bond and pnicogen bond are commonly responsible for the stability of other complexes. In **3b**, the $LP_P \rightarrow BD^*_{H-O}$ orbital interaction corresponding to the H···P hydrogen bond has dominant contribution to its stability, although its perturbation energy shows a different change from that in **3a**.

3.5. NCI plots

Non-covalent interactions (NCI) index proposed by Yang's group [65,66] was demonstrated to provide an evidence for the existence of van der Waals interactions, hydrogen bonds, and repulsive steric clashes. Additionally, this method was also used to study halogen bonds [67,68], and it was found that halogen bonds exhibit similar character of NCI with hydrogen bonds. Furthermore, the strengths of noncovalent interactions are also measured with this method [68]. Here it is applied to study pnicogen bonds, besides hydrogen and halogen bonds. NCI involves the reduced density gradient (*RDG*) and the electron density (ρ). *RDG* is defined as:

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$$RDG = \frac{1}{2(3\pi^2)^{1/3}} \frac{|\nabla\rho|}{\rho^{4/3}}$$

Fig. 4 shows the graph of *RDG* vs sign(λ_2)ρ, where λ_2 is the second eigenvalue in Hessian matrix, for HOBr:PH_2Br complexes. Other complexes exhibit a similar feature. In Fig. 4a, two spikes with low density, low-gradient are observed: one corresponds to pnicogen bonding and the other to the H···Br interaction, which is not detected in the AIM maps. This means that NCI is a reliable method for detecting weak interactions with similar strength to van der Waals interactions, and confirms the conclusion that NCI is able to recover the features of AIM in real space, but NCI features are present even in the absence of critical points [69]. Furthermore, the pnicogen bond has a ρ value well in the range of typical hydrogen bond interactions (ρ ≈ 0.02 au), indicating more electrostatic interaction, while the H···Br weak interaction shows characteristics of a more dispersion interaction (ρ < 0.01 au) [70]. Additionally, there are two repulsive interactions with a positive value of sign (λ_2)ρ (Fig. 4a). In Fig. 4c, there is a negative value of sign (λ_2)ρ, which is indicative of a Br···P halogen bond, also spotted in a blue disk in gradient isosurfaces (Fig. 4d). Clearly, the Br···P halogen bond in **2** of HOBr:PH_2Br is stronger than the pnicogen bond in **1**. There is repulsive interaction between the lone pair electrons of Br and P atoms, which is marked by a positive value of sign (λ_2)ρ (Fig. 4c). This feature is like that of halogen bond in F_3CX-trimethyl phosphine (X = Cl, Br, and I) [67]. **3** exhibits a similar characteristic of NCI to **1** with two negative spikes. Both negative spikes appear in the same region with low density, which indicates that the P···Br pnicogen bond can compete with the P···H hydrogen bond.

4. Conclusions

Ab initio MP2/aug-cc-pVTZ calculations have been performed to identify local minima

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on the potential surfaces of the complexes formed by HOX (X = Cl, Br, and I) and phosphine derivatives (PH_2Y, Y = H, F, Cl, Br, CH_3, NH_2, OH, and NO_2), to characterize the types of interactions in these complexes, and to evaluate their interaction strengths. The following conclusions have been summarized.

(1) Four equilibrium structures (**1**, **2**, **3a**, and **3b**) have been found on the potential surfaces of most complexes except HOBr:PH_2Y (Y = H, CH_3, NH_2, and OH), for which only **2, 3a,** and **3b** are obtained.

(2) The structure **1** is stabilized by an O···P pnicogen bond and a secondary H···X interaction. Its stability can be regulated through the substituents X and Y, and it becomes more stable in order of X = Cl < Br < I and for the stronger electron-withdrawing group Y.

(3) The X···P halogen bond is responsible for the stability of **2**. The halogen atom as the halogen donor has a great effect on its stability and the effect is insignificant when it acts as the substituent in the halogen acceptor.

(4) The structure **3a** is stabilized by a H···P hydrogen bond and a P···X pnicogen bond. According to the orbital interactions, the former has a bigger contribution than the latter in the complexes of HOCl:PH_2F and HOBr:PH_2OH, the latter contribution is negligible in the complexes of HOBr:PH_2Y (Y = H, CH_3, NH_2), both types of orbital interactions have almost equal contribution in HOBr:PH_2F, and the former shows a smaller contribution than the latter in the complexes of HOI:PH_2F, HOBr:PH_2Cl, HOBr:PH_2Br, and HOBr:PH_2NO_2.

(5). The H···P hydrogen bond also plays a dominant role in **3b**, which is more stable than **3a**.

Acknowledgements

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References

[1]. S. Scheiner, Hydrogen Bonding: A Theoretical Perspecitive, Oxford University Press, New York, 1997.

[2]. G.A. Jeffrey, An Introduction to Hydrogen Bonding, Oxford University Press, New York, 1997.

[3]. R. Custelcean, J. E. Jackson, Chem. Rev. 101 (2001) 1963–1980.

[4]. P. Politzer, J. S. Murray, ChemPhysChem 14 (2013) 278–294.

[5]. S. Scheiner, Acc. Chem. Res. 46 (2013) 280–288.

[6]. E. Arunan, G.R. Desiraju, R.A. Klein, J. Sadlej, S. Scheiner, I. Alkorta, D.C. Clary, R.H. Crabtree, J.J. Dannenberg, P. Hobza, H.G. Kjaergaard, A.C. Legon, B. Mennucci, D.J. Nesbitt, Pure Appl. Chem. 83 (2011) 1637–1641.

[7]. L.C. Gilday, T. Lang, A. Caballero, P.J. Costa, V. Felix, P.D. Beer, Angew Chem. Int. Ed. 52 (2013) 4356–4360.

[8]. A.V. Jentzsch, S. Matile, J. Am. Chem. Soc. 135 (2013) 5302−5303.

[9]. H.R. Khavasi, A.A. Tehrani, Inorg. Chem. 52 (2013) 2891−2905.

[10]. J.E. Ormond-Prout, P. Smart, L. Brammer, Cryst. Growth Des. 12 (2012) 205−216.

[11]. H.S. El-Sheshtawy, B.S. Bassil, K.I. Assaf, U. Kortz, W.M. Nau, J. Am. Chem. Soc. 134 (2012) 19935−19941.

[12]. L. Meazza, J.A. Foster, K. Fucke, P. Metrangolo, G. Resnati, J.W. Steed, Nat. Chem. 5 (2013) 42–47.

[13]. M.G. Sarwar, B. Dragisic, S. Sagoo, M.S. Taylor, Angew. Chem. Int. Ed. 49 (2010) 1674–1677.

[14]. P. Metrangolo, J.S. Murray, T. Pilati, P. Politzer, G. Resnati, G. Terraneo, CrystEngComm 13 (2011) 6593–6596.

[15]. P. Auffinger, F.A. Hays, E. Westhof, P.S. Ho, Proc. Natl. Acad. Sci. SUA 101 (2004)

ACCEPTED MANUSCRIPT

16789–16794.

[16]. G.R. Desiraju, P.S. Ho, L. Kloo, A.C. Legon, R. Marquardt, P. Metrangolo, P. Politzer, G. Resnati, K. Rissanen, Pure Appl. Chem. 85 (2013) 1711–1713.

[17]. U. Adhikari, S. Scheiner, J. Phys. Chem. A 116 (2012) 3487–3497.

[18]. J.E. Del Bene, I. Alkorta, G. Sánchez-Sanz, J. Elguero, J. Phys. Chem. A 115 (2011) 13724–13731.

[19]. S. Scheiner, J. Phys. Chem. A 115 (2011) 11202–11209.

[20]. S. Scheiner, Chem. Phys. 387 (2011) 79–84.

[21]. S. Scheiner, J. Chem. Phys. 134 (2011) 094315–094323.

[22]. M. Solimannejad, M. Gharabaghi, S. Scheiner, J. Chem. Phys. 134 (2011) 024312–043217.

[23]. J.E. Del Bene, I. Alkorta, G. Sánchez-Sanz, J. Elguero, Chem. Phys. Lett. 534 (2012) 14–18.

[24]. J.E. Del Bene, I. Alkorta, G. Sánchez-Sanz, J. Elguero, J. Phys. Chem. A 116 (2012) 3056–3060.

[25]. J.E. Del Bene, I. Alkorta, G. Sánchez-Sanz, J. Elguero, Chem. Phys. Lett. 512 (2011) 184–187.

[26]. S. Scheiner, Int. J. Quantum Chem. 113 (2013) 1609–1620.

[27]. S. Scheiner, CrystEngComm 15 (2013) 3119–3124.

[28]. S.J. Grabowski, I. Alkorta, J. Elguero, J. Phys. Chem. A 117 (2013) 3243–3251.

[29]. I. Alkorta, G. Sánchez-Sanz, J. Elguero, J.E. Del Bene, J. Phys. Chem. A 117 (2013) 183–191.

[30]. I. Alkorta, J. Elguero, J.E. Del Bene, J. Phys. Chem. A 117 (2013) 4981–4987.

[31]. J.E. Del Bene, I. Alkorta, G. Sánchez-Sanz, J. Elguero, J. Phys. Chem. A 117 (2013) 3133–3141.

[32]. Q.Z. Li, R. Li, X.F. Liu, W.Z. Li, J.B. Cheng, ChemPhysChem 13 (2012) 1205–1212.

[33]. Q.Z. Li, R. Li, X.F. Liu, W.Z. Li, J.B. Cheng, J. Phys. Chem. A 116 (2012) 2547–2553.

[34]. S. Zahn, R. Frank, E. Hey-Hawkins, B. Kirchner, Chem. Eur. J. 17 (2011) 6034–6038.

[35]. A. Bauza, D. Quinonero, P.M. Deyaa, A. Frontera, Phys. Chem. Chem. Phys. 14 (2012) 14061–14066.

ACCEPTED MANUSCRIPT

[36]. J.S. Francisco, S.P. Sander, J. Chem. Phys. 99 (1993) 6219–6220.

[37]. M.J. Molina, F.S. Rowland, Nature 249 (1974) 810–812.

[38]. F.S. Rowland, M.J. Molina, Rev. Geophys. Space Phys. 13 (1975) 1–35.

[39]. E. Thomas, Infect. Immun. 23 (1979) 522–531.

[40]. Z.F. Zhang, J. Shen, N.Z. Jin, L.P. Chen, Z.Y. Yang, Comput. Theor. Chem. 999 (2012) 48–54.

[41]. J.J. Panek, S. Berski, Chem. Phys .Lett. 467 (2008) 41–45.

[42]. H. Roohi, A. Nowroozi, F. Eshghi, Int. J. Quantum Chem. 110 (2010) 1489–1499.

[43]. M. Solimannejad, I. Alkorta, Chem. Phys. Lett. 454 (2008) 201–206.

[44]. K. Yuan, Y.Z. Liu, Y.C. Zhu, J. Zhang, J.Y. Zhang, Acta Chin. Sin. 67 (2009) 499–506.

[45]. K. Yuan, Y.Z. Liu, L.L. Lue, W.C. Ma, Acta Chin. Sin. 24 (2008) 1257–1263.

[46]. K. Yuan, Y.Z. Liu, W.C. Ma, H.A. Tang, Y.C. Zhu, J. Zhang, Chi. J. Chem. 27 (2009) 900–906.

[47]. Z.F. Li, X.Y. Wang, H.A. Tang, Y.C. Zhu, H.Y. Li, Chem. J. Chin. Univ. 30 (2009) 92–95.

[48]. Z.F. Li, H.Y. Li, Y.Z. Liu, X.N. Shi, H.A. Tang, Chin. Chem. Bull. 54 (2009) 3014–3022.

[49]. I. Alkorta, F. Blanco, M. Solimannejad, J. Elguero, J. Phys. Chem. A 112 (2008) 10856–10863.

[50]. Q.Z. Li, X.S. Xu, T. Liu, B. Jing, W.Z. Li, J.B. Cheng, B.A. Gong, J.Z. Sun, Phys. Chem. Chem. Phys. 12 (2010) 6837–6843.

[51]. Q.Z. Li. B. Jing, R. Li, Z.B. Liu, W.Z. Li, F. Luan, J.B. Cheng, B.A. Gong, J.Z. Sun, Phys. Chem. Chem. Phys. 13 (2011) 2266–2271.

[52]. Li, Q.Z. Zhao, J.L. Jing, B. Li, R. Li, W.Z. Cheng, J.B. J. Comput. Chem. 2011, 32, 2432–2440.

[53]. F. Blanco, I. Alkorta, M. Solimannejad, J. Elguero, J. Phys. Chem. A 113 (2009) 3237–3244.

[54]. Q. Zhao, D. Feng, Y. Sun, J. Hao, Z. Cai, J. Mol. Model. 17 (2011) 1935–1939.

[55]. A. Zabaradsti, A. Kakanejadifard, M. Ghasemian, Comput. Theor. Chem. 989 (2012) 1–6.

ACCEPTED MANUSCRIPT

[56]. M.J. Frisch, G.W. Trucks, H.B. Schlegel, et al., Gaussian 09 (Revision A.02), Gaussian, Inc., Wallingford, CT, 2009.

[57]. S.F. Boys, F. Bernardi, Mol. Phys. 19 (1970) 553–566.

[58]. F.A. Bulat, A. Toro-Labbé, T. Brinck, J.S. Murray, P. Politzer, J. Mol. Model. 16 (2010) 1679–691.

[59]. A.E. Reed, L.A. Curtiss, F. Weinhold, Chem. Rev. 88 (1988) 89–926.

[60]. R.F.W. Bader, Atoms in Molecules: A Quantum Theory, Oxford University Press, Oxford, UK, 1990.

[61]. T. Lu, F. Chen, J. Comput. Chem. 33 (2012) 580–592.

[62]. W. Humphrey, A. Dalke, K. Schulten, J. Mol. Graphics 14 (1996) 33–38.

[63]. J.B. Cheng, R. Li, Q.Z. Li, B. Jing, Z.B. Liu, W.Z. Li, B.A. Gong, J.Z. Sun, J. Phys. Chem. A 114 (2010) 10320–10325.

[64]. A.J. Stone, J. Am. Chem. Soc. 135 (2013) 7005–7009.

[65]. J. Contreras-García, E.R. Johnson, S. Keinan, R. Chaudret, J.P. Piquemal, D.N. Beratan, W.T. Yang, J. Chem. Theory Comput. 7 (2011) 625–632.

[66]. E. R. Johnson, S. Keinan, P. Mori-Sanchez, J. Contreras-Garcia, A. J. Cohen, W.T. Yang, J. Am. Chem. Soc. 132 (2010) 6498–6506.

[67]. B. Pinter, N. Nagels, W.A. Herrebout, F.D. Proft, Chem. Eur. J. 19 (2013) 519–530.

[68]. S. Kozuch, J. M. L. Martin, J. Chem. Theory Comput. 9 (2013) 1918−1931.

[69]. J. Contreras-García, W. T. Yang, E. R. Johnson, J. Phys. Chem. A 115 (2011) 12983–12990.

[70]. K. E. Riley, P. Hobza, J. Chem. Theory Comput. 4 (2008) 232−242.

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Table 1

Maximal and minimal electrostatic potentials (V, eV) on the pnicogen atoms in the phosphine derivatives (PH_2Y, Y = H, F, Cl, Br, CH_3, NH_2, OH, and NO_2) and on the H, O, and X atoms in HOX as well as MK electrostatic potential charge (Q_Y, e) on the Y of PH_2Y at the MP2/aug-cc-pVTZ level.

PH_2Y	$V_{max,P}$	$V_{min,P}$	HOX	$V_{max,H}$	$V_{max,X}$	$V_{min,X}$	$V_{min,O}$
Y = H	0.0203	-0.0257	X = Cl	0.0893	0.0381	-0.0149	-0.0300
Y = F	0.0604	-0.0190	X = Br	0.0857	0.0499	-0.0125	-0.0319
Y = Cl	0.0523	-0.0147	X = I	0.0810	0.0631	-0.0096	-0.0350
Y = Br	0.0487	-0.0136	PH_2Y	Q_Y			
Y = CH_3	0.0128	-0.0342	Y = H	0.059			
Y = NH_2	0.0264	-0.0311	Y = CH_3	0.096			
Y = OH	0.0402	-0.0240	Y = NH_2	-0.062			
Y = NO_2	0.0702	0.0005	Y = OH	-0.061			

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Table 2

Interaction energy corrected for BSSE (ΔE, kJ/mol) in HOX:PH_2Y (X = Cl, Br, I; Y = H, F, Cl, Br, CH_3, NH_2, OH, and NO_2) at the MP2/aug-cc-pVTZ level.

	ΔE_1	ΔE_2	ΔE_{3a}	ΔE_{3b}
HOCl:PH_2F	-15.34	-10.22	-20.84	---
HOBr:PH_2F	-16.57	-17.19	-21.39	---
HOI:PH_2F	-17.96	-28.81	-21.89	---
HOBr:PH_2Cl	-14.88	-17.93	-18.99	---
HOBr:PH_2Br	-14.26	-17.42	-18.34	---
HOBr:PH_2H	---	-19.07	-20.32	---
HOBr:PH_2CH_3	---	-29.24	-25.81	-27.59
HOBr:PH_2NH_2	---	-29.40	-24.85	-28.18
HOBr:PH_2OH	---	-28.32	-22.63	-27.90
HOBr:PH_2NO_2	-17.69	-12.24	-14.32	---

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Table 3

Binding distances (R, Å) in the complexes at the MP2/aug-cc-pVTZ level.

	R_1	R_2	R_3	R_4	R_5	R_6
HOCl:PH_2F	2.731	3.097	2.448	3.116	---	---
HOBr:PH_2F	2.699	2.720	2.462	3.160	---	---
HOI:PH_2F	2.687	2.813	2.513	3.268	---	---
HOBr:PH_2Cl	2.784	2.848	2.469	3.250	---	---
HOBr:PH_2Br	2.789	2.790	2.464	3.262	---	---
HOBr:PH_2H	---	2.896	2.380	3.689	---	---
HOBr:PH_2CH_3	---	2.663	2.319	3.705	2.319	3.300
HOBr:PH_2NH_2	---	2.635	2.341	3.511	2.325	2.877
HOBr:PH_2OH	---	2.632	2.391	3.331	2.350	2.601
HOBr:PH_2NO_2	2.738	3.018	2.577	3.208	---	---

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Table 4

Frequency shifts ($\Delta\nu$, cm^{-1}) of H-O(1), X-O(2), and P-Y(3) stretch vibrations in the complexes at the MP2/aug-cc-pVTZ level.

	$\Delta\nu_{1,\mathbf{1}}$	$\Delta\nu_{1,\mathbf{2}}$	$\Delta\nu_{1,\mathbf{3a}}$	$\Delta\nu_{2,\mathbf{1}}$	$\Delta\nu_{2,\mathbf{2}}$	$\Delta\nu_{2,\mathbf{3a}}$	$\Delta\nu_{3,\mathbf{1}}$	$\Delta\nu_{3,\mathbf{2}}$	$\Delta\nu_{3,\mathbf{3a}}$
HOCl:PH_2F	-21.2	-3.7	-204.8	-5.1	-44.0	3.2	-17.0	9.0	9.2
HOBr:PH_2F	-16.5	-4.0	-196.1	-3.6	-97.3	5.3	-21.0	18.1	4.5
HOI:PH_2F	-14.2	12.9	-179.2	-1.8	-68.6	8.2	-24.9	31.7	-3.4
HOBr:PH_2Cl	-17.1	-0.7	-182.9	-3.0	-79.2	4.8	-15.7	16.5	2.7
HOBr:PH_2Br	-17.7	-2.6	-180.6	-3.5	-90.1	4.4	-10.4	13.7	5.2
HOBr:PH_2H	---	2.5	-227.4	---	-76.4	6.1	---	16.9	20.6
HOBr:PH_2CH_3	-318.7[a]	-1.3	-311.2	6.5[a]	-150.9	7.0	14.6[a]	26.8	14.7
HOBr:PH_2NH_2	-322.0[a]	-2.8	-292.8	7.7[a]	-152.4	6.6	42.1[a]	48.9	12.9
HOBr:PH_2OH	-307.0[a]	16.7	-253.2	6.6[a]	-149.2	5.3	29.1[a]	48.1	15.4
HOBr:PH_2NO_2	-17.8	-4.4	-121.2	-3.7	-33.8	4.8	-4.5	4.0	0.9

[a] The data are from **3b**.

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Table 5

Second-order perturbation energy (E, kJ/mol) in the complexes at the HF/aug-cc-pVTZ level.

	$E_{1,\mathbf{1}}$	$E_{2,\mathbf{1}}$	$E_{1,\mathbf{2}}$	$E_{2,\mathbf{2}}$	$E_{1,\mathbf{3a}}$	$E_{2,\mathbf{3a}}$	$E_{1,\mathbf{3b}}$	$E_{2,\mathbf{3b}}$
HOCl:PH_2F	22.11	1.92	32.48	3.05	38.25	25.87	---	---
HOBr:PH_2F	24.41	2.47	153.20	36.58	34.53	35.40	---	---
HOI:PH_2F	26.17	2.05	235.71	20.65	25.75	44.01	---	---
HOBr:PH_2Cl	24.03	1.29	111.73	7.90	27.96	31.73	---	---
HOBr:PH_2Br	25.46	1.21	119.67	24.41	26.84	32.27	---	---
HOBr:PH_2H	---	---	107.09	4.39	56.85	5.77	---	---
HOBr:PH_2CH_3	---	---	243.78	4.43	84.14	4.89	81.13	0.79
HOBr:PH_2NH_2	---	---	278.56	6.27	76.54	10.78	84.94	4.47
HOBr:PH_2OH	---	---	262.00	10.49	55.43	20.44	78.54	17.89
HOBr:PH_2NO_2	27.75	1.09	53.29	4.56	15.05	39.17	---	---

Note: E_1 corresponds to the orbital interaction of $LP_O \rightarrow BD^*_{P\text{-}Y}$ in **1**, $LP_P \rightarrow BD^*_{X\text{-}O}$ in **2**, and $LP_P \rightarrow BD^*_{H\text{-}O}$ in **3a** and **3b**. E_2 corresponds to the orbital interaction $LP_P \rightarrow BD^*_{X\text{-}O}$ in **1**, $LP_X \rightarrow BD^*_{P\text{-}Y}$ in **2** and **3a**, and $LP_{Br} \rightarrow BD^*_{H\text{-}Y}$ in **3b**.

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Figure captions

Fig. 1 Molecular electrostatic potentials of PH_2F and HOCl. Color ranges, in eV, are: red, greater than 0.03; yellow, between 0.03 and 0; green, between 0 and -0.01; blue, less than -0.01.

Fig. 2 Structures of **1**, **2**, **3a**, and **3b** of HOX:PH_2Y (X = Cl, Br, and I; Y = H, F, Cl, Br, CH_3, NH_2, OH, and NO_2).

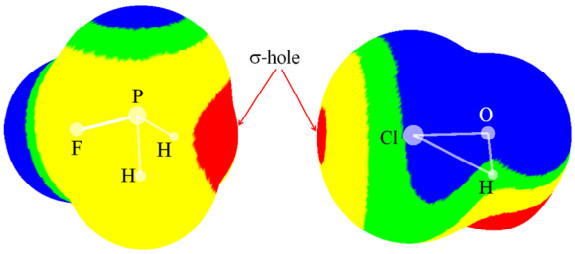
Fig. 3 NBO orbital maps in **1** (top), **2** (middle), and **3a** (down) of HOBr:PH_2Br.

Fig. 4 Plots of the reduced density gradient (RDG) versus the electron density multiplied by the sign of the second Hessian eigenvalue in **1** (top), **2** (middle), and **3a** (down) of HOBr:PH_2Br.

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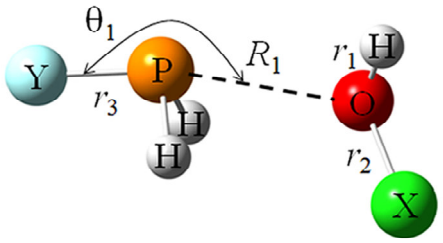
Fig. 1



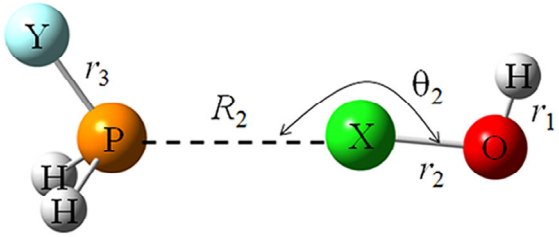
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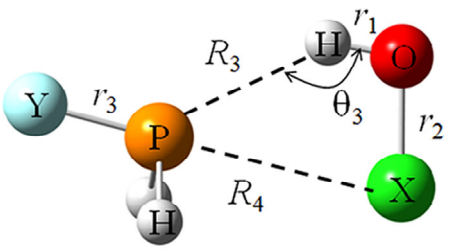
Fig. 2



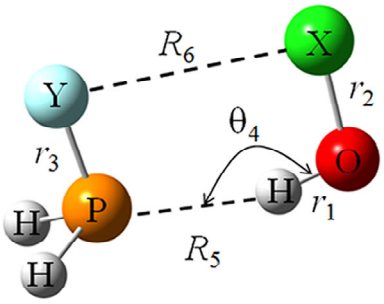
1



2



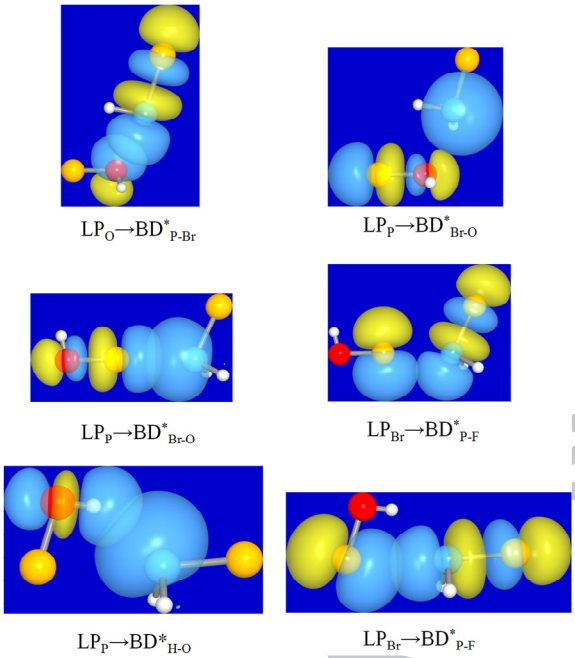
3a



3b

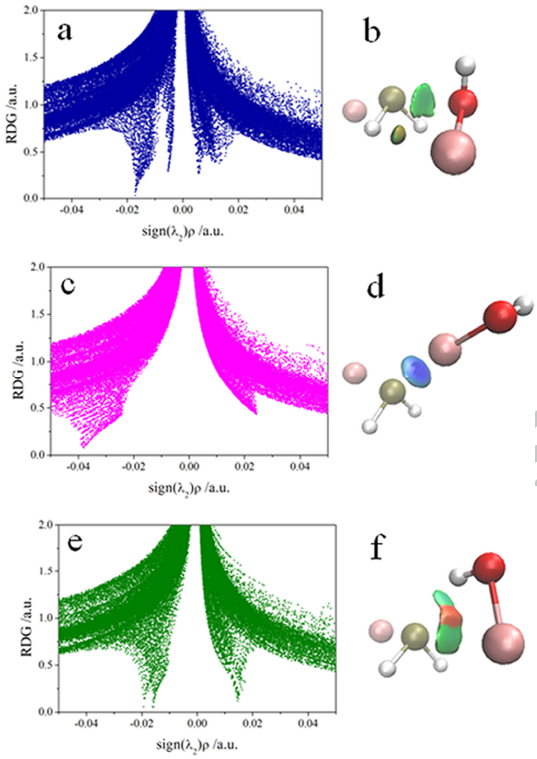
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Fig. 3

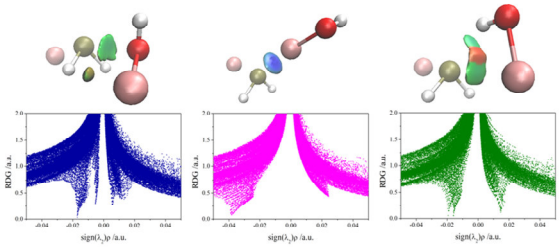


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Fig. 4



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- Three types of complexes were found for the complexes of hypohalous acids and phosphine derivatives.
- There is competition among hydrogen bond, halogen bond, and pnicogen bond.
- This competition is related to the nature of hypohalous acids and phosphine derivatives.

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